

$$z_{\text{cr}}^{\text{Hz}} = \left(\sum_{i=1}^{N-1} \mathbf{1}\{s_{i-1}s_i < 0\} \right) \frac{f_s}{2(N-1)}$$

where $s = (s_0, \dots, s_{N-1})$ is the discrete signal of length N and f_s is the sampling frequency.